
A Reproducible Benchmark for Earned vs Enforced Measurement Consistency in Dynamic Interferometric Imaging

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Abstract

1 Structured inverse problems are often evaluated on coefficients that were either
2 directly observed or explicitly restored by a data-consistency layer, which makes it
3 difficult to tell whether a model truly learned to recover unmeasured information.
4 We propose a reproducible benchmark for *earned* versus *enforced* measurement
5 consistency under structured Fourier missingness. Dynamic VLBI is the motivating
6 case because it is extreme: sparse complex-valued measurements, non-Cartesian
7 Fourier coverage, and time variability [Event Horizon Telescope Collaboration,
8 2019a, next Generation Event Horizon Telescope, 2026]. The benchmark fixes
9 deterministic support-target manifests, three holdout families, release-aware public
10 validation, and paired-bootstrap reporting. On the released synthetic breadth matrix,
11 the earned-consistency reference model EMC is strongest in all 12 family-support
12 cells, while public EHT validation reveals release-dependent transfer limitations
13 that become partly but not uniformly smaller under test-time optimization. Those
14 mixed public outcomes are useful because they expose the residual synthetic-to-real
15 gap rather than hiding it behind pooled averages. The protocol is not specific to
16 VLBI; it applies directly to time-varying MRI settings in which extrapolation to
17 unobserved Fourier coefficients is equally central.

18 1 The benchmark question

19 The central question is simple: can a reconstruction method recover structured Fourier coefficients
20 that were *never shown* to the model and *never restored* by a support-set data-consistency operator?
21 We call success on that task *earned* measurement consistency. By contrast, *enforced* consistency
22 arises when observed coefficients are reinserted explicitly, so low error on those same coefficients
23 is no longer evidence of learned extrapolation. This distinction matters across structured inverse
24 problems, but dynamic VLBI makes it unusually sharp because sparse time-varying baselines leave
25 little room to hide evaluation leakage [Event Horizon Telescope Collaboration, 2019b, 2022].

26 2 Protocol

27 Our protocol fixes deterministic support-target manifests under three structured missingness families:
28 baseline-track blocks, scan-segment blocks, and station dropout. Each family is evaluated at support
29 fractions of 20, 40, 60, and 80 per cent on a common 64×64 grid. The benchmark release stores split
30 manifests, configuration manifests, results manifests, seeds, paired-bootstrap summaries, public-EHT
31 release metadata, and figure/table generation scripts. Released public validation spans four official
32 calibrated-data products: M87 2017, M87 2018, 3C279 2017, and Centaurus A 2017.

Table 1: Synthetic benchmark summary.

Result	Value
Released breadth matrix	EMC best learned model in 12/12 family-support cells
Baseline-track 20% support	63.3% lower held-out visibility-error power than residual refinement
Five-seed baseline-track study	Residual refinement strongest mean learned comparator in 4/4 support fractions
Conformal UQ on EMC	Mean empirical 90% coverage = 85.8%; mean interval width = 0.168

Table 2: Public EHT baseline-track release summary with TTO. Positive uplift means lower held-out visibility RMSE than plain EMC.

Release	EMC-TTO uplift	Release-level outcome
M87 2017	+0.0696	Improved, but Tikhonov remains strongest mean model
M87 2018	-0.0833	Degraded under adaptation
3C279 2017	+0.2275	Best release-level transfer case; EMC-TTO wins
Centaurus A 2017	-0.1862	Degraded under adaptation

3 Key results

Table 1 summarizes the synthetic benchmark signal. EMC is the strongest learned method in all 12 cells of the released breadth matrix, and on the baseline-track 20 per cent slice it reduces held-out visibility-error power by 63.3 per cent relative to residual refinement, the strongest enforced learned comparator in that slice. A bounded five-seed follow-up narrows the method claim, but it does not weaken the benchmark claim: the protocol is informative precisely because it separates favorable released breadth from stricter robustness.

Table 2 shows the public-EHT reality check. Test-time optimization (TTO) helps on M87 2017 and strongly on 3C279 2017, but degrades M87 2018 and Centaurus A 2017. The public story is therefore benchmark-strengthening rather than method-conclusive: structured hold-out evaluation continues to separate regimes after leaving the synthetic domain. The add-on cycle also introduces a bounded Diffusion Posterior Sampling (DPS) comparator [Chung et al., 2023], which gives an earned-consistency diffusion baseline without access to held-out target measurements.

4 Cross-domain applicability

The protocol is not unique to VLBI. Cardiac cine MRI has the same mathematical structure: a time-varying source observed through undersampled Fourier measurements with structured missingness across time. Methods that enforce k-space agreement at observed frequencies need not generalize to unobserved frequencies across cardiac phases. FastMRI-style cine benchmarks could therefore test the same earned-versus-enforced protocol in a domain with image-domain ground truth and higher clinical stakes [Zbontar et al., 2018, Cui et al., 2024]. The same logic also connects naturally to recent diffusion-based interferometric imaging work in astronomy [Lai et al., 2025]. In that sense, VLBI is the extreme structured-inverse-problem case, not the only one.

5 Conclusion

The benchmark release makes one point reproducibly: agreement on observed coefficients is not enough. In dynamic interferometric imaging, the scientifically relevant question is whether a method can recover unobserved coefficients under structured missingness, and whether that ability survives contact with released public measurements. The same benchmark logic extends naturally to other time-varying Fourier inverse problems, including cardiac cine MRI. We release the manifests, artifact builders, theory note, and public-EHT validation suite to support that broader comparison agenda.

EMC benchmark: held-out measurement recovery across structured holdout families

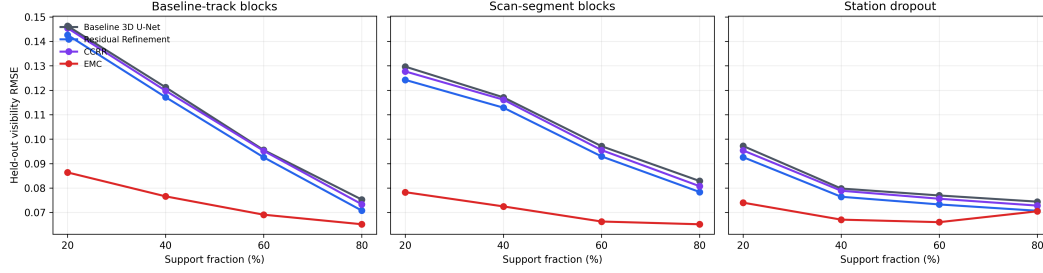


Figure 1: Support-fraction curves for the released synthetic benchmark. The main signal is not that one architecture always wins, but that earned-vs-enforced evaluation yields a stable and reproducible separation across structured missingness families.

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